Highlights

BowTie Risk Analysis of the Valve Train in a High-Performance VW AP 1.8 Cylinder Head Operating at 10,000 rpm

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- Quantitative-coupled BowTie for high-rpm valve-train risk
- Top event: loss of valve–cam follower contact
- Outer-spring coil-bind margin 0.78 mm vs project 1.5 mm
- Inner-spring Goodman FoS 0.68 fails in 100% of MC trials
- Monte Carlo refinement under ASTM A877/A877M tolerances

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Abstract

The valve train is load-critical when a spark-ignition engine is pushed to high rotational speeds: at 10 000 rpm inertial demand, spring dynamics, and head thermal management become tightly coupled and small departures from the design envelope can cascade into valve–piston collision. This work distils the forensic root causes reported in recent valve-train failure case studies into a structured, deterministic preventive procedure: a quantitative-coupled BowTie for design-stage reliability assessment of high-rpm valve trains, applied to a high-performance VW AP 1.8 head at 10 000 rpm. Threats, preventive barriers, consequences, and mitigative barriers are mapped around the top event "loss of follower contact between valve and cam"; each preventive barrier is linked to a deterministic indicator with a literature-anchored pass criterion, derived from the analytical sizing of the intake valve and the dual Cr–Si springs. All four dynamic-ratio indicators meet the project criterion ($R \geq 1.20$, worst case 1.343), but two concurrent at-risk barriers are identified: the outer-spring coil-bind margin ($M_{\text{out}} = 0.78$ mm vs 1.5 mm) and the inner-spring Goodman factor of safety (FoS) ($n_{f,\text{in}} \approx 0.68$ vs 1). The two findings affect different components and demand independent design interventions. A Monte Carlo refinement under ASTM A877/A877M tol-

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erances ($N = 2 \times 10^5$ trials) confirms the inner-spring Goodman deficit in 100 % of the manufactured population and re-classifies the inner-spring coil-bind margin from active to marginal ($P[\text{fail}] = 16.2\,\%$). A construct-validity benchmark against four published valve-train failure investigations captures three cases within the framework and identifies the fourth as a documented barrier-set extension. A test-rig validation programme with statistical acceptance criteria closes the empirical gap.

Keywords: BowTie analysis, Valve train, Risk assessment, Internal combustion engine, Mechanical reliability

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1. Introduction

Spark-ignition (SI) engines remain technologically relevant in light automotive applications, particularly in markets where cost pressure, mechanical robustness, an established refuelling network, and a gradual transition to lower-impact energy solutions coexist [1, 2]. Within this landscape, the search for higher specific output and thermal efficiency does not depend solely on electronic calibration or fuel selection: the geometric quality of the intake system, of the combustion chamber, and of the valve train, together with thermal management of the head, governs the practical envelope of operation.

In high-performance configurations, where elevated rotational speed, increased thermal loading, and stricter operational stability are required simultaneously, two risks acquire a critical role: abnormal combustion (knock) and valve-train failure. Both phenomena compromise reliability, shrink the operational safety margin, and may evolve from a local component issue into catastrophic damage to the engine. The present work focuses exclusively on the second of these two risks. The companion problem of knock at high compression ratio is addressed elsewhere and is out of scope here.

The Volkswagen AP 1.8 platform was adopted as the reference engine for two reasons. First, it occupies an established position in the Brazilian automotive market, with a wide installed base and broad availability of geometric data [3, § 1.6]. Second, in the context of a master's research project investigating a high-performance cylinder head for the same platform, a specific operating regime of 10 000 rpm was adopted as the design target. At that rotational speed the inertial demand placed on the valve–cam contact pair is no longer a local component concern: the dynamic stability of the moving valve mass, the residual coil-bind clearance, the natural frequency of the spring relative to cam harmonics, and the fatigue resistance of the spring all become coupled, and a single insufficient margin can propagate into a valve–piston collision.

The conventional Failure Mode and Effects Analysis (FMEA) approach addresses such situations by ranking failure modes through the multiplicative Risk Priority Number (RPN). This approach has known limitations: subjectivity in the assignment of severity, occurrence, and detectability scales; loss of resolution when multiple modes share a similar RPN; and limited visibility of the causal chain that links threats to consequences via barriers. The BowTie methodology addresses these limitations by organising the analysis around a top event, with threats and preventive barriers on the left side, and consequences and mitigative barriers on the right side. This structure makes the role of each barrier explicit and supports communication between design, operation, and reliability domains.

This paper proposes a quantitative-coupled BowTie analysis of the valve train of a high-performance cylinder head operating at 10 000 rpm in the VW AP 1.8 engine. The qualitative bow diagram is enriched with a quantitative layer derived from the analytical sizing of the intake valve and of the valve springs. Each preventive barrier is associated with a quantitative pass criterion (natural-frequency margin, coil-bind margin, fatigue factor of safety) so that the BowTie diagram becomes both a communication artefact and a design check.

The contributions of this paper are summarised as follows:

1. A BowTie diagram of the valve train of the AP 1.8 cylinder head at 10 000 rpm, with the top event "loss of follower contact between valve and cam" and the associated set of threats, preventive barriers, consequences, and mitigative barriers.
2. A quantitative coupling layer that translates each preventive barrier into a measurable indicator with a literature-anchored pass criterion, derived from the analytical sizing of the valve and the spring.
3. Identification of the critical preventive barrier for the operating envelope adopted, with a comparison against a conventional FMEA/RPN ranking of the same failure modes.
4. A reproducible methodology that can be transposed to other high-performance cylinder heads operating at high rotational speed.

Positioning of the contribution. This work operates at the interface between forensic failure analysis and design-stage prevention: it takes the root causes reported in the recent valve-train failure literature [4, 5, 6, 7] as the empirical basis and distils them into a structured, deterministic preventive procedure usable before a prototype is built. The relevant failure mechanisms of the high-rpm valve train (valve float, bounce, approach to coil bind, surge resonance, fatigue) are reorganised around the BowTie methodology, and the analytical sizing of the valve and the springs is mapped, barrier by barrier, onto deterministic indicators with literature-anchored pass criteria. The contribution is therefore best characterised as *forensic-informed prevention*: the forensic content is borrowed faithfully from peer-reviewed case studies; the original step is the translation of those root causes into a reproducible, indicator-driven barrier check applicable at the design stage. Quantitative extensions of the BowTie diagram have been previously proposed (8, via Bayesian-network mapping; LOPA, fault-tree-coupled BowTie, and DyPASI in process safety), and the present work does not claim a new general method; the explicit mapping of each preventive barrier to a deterministic indicator derived from the engineering sizing of the intake valve and the dual valve springs of the AP 1.8 head — with the identification of the resulting critical barriers under the operating envelope adopted — is the original contribution. The procedure is benchmarked against the same four documented failure cases: three of the four root causes are captured by the deterministic indicator layer (*coil bind / fatigue / dynamic loss of contact*), and the fourth case [7] defines the boundary of the current barrier set and motivates a documented extension to the lubrication-regime barrier (see Section 5.3). An empirical verification on a dedicated test rig with statistical acceptance criteria is identified as the priority direction for future work, and a concrete validation roadmap is specified in Section 6. The claim that no published study applies the BowTie methodology to the valve train of a high-rpm spark-ignition cylinder head is supported by a search of the Scopus and Web of Science databases conducted in April 2026 with the query strings `("BowTie" OR "bow-tie") AND ("valve train" OR "valve-train" OR "valve spring") AND ("spark ignition" OR "SI engine" OR "high rpm")`, which returned no relevant hits.

Section 2 reviews the relevant failure modes of the valve train at high rpm and the BowTie methodology. Section 3 describes the system under study, the operational premises, the qualitative BowTie construction, and the quantitative-coupling procedure. Section 4 reports the populated BowTie diagram, the values of the quantitative indicators for the AP 1.8 case, and the qualitative validation of the procedure against four published valve-train failure cases. Section 5 discusses the implications for cylinder-head design, compares the proposed approach with FMEA/RPN, and states the limitations of the present analytical study. Section 6 specifies the experimental validation roadmap and the candidate barrier extensions, and Section 7 closes the paper. The work complements the post-failure case-study tradition of Vélez Godiño et al. [4], Soffritti et al. [5], Xing et al. [6], Zhao et al. [7], which informed the present formulation, by translating their lessons into a structured prospective barrier-evaluation procedure applicable at the design stage.

2. Background

2.1. Valve-train failure modes at high rotational speed

At elevated rotational speeds the valve train of an SI engine moves from a quasi-static loading regime, in which the valve simply tracks the cam profile, into an inertia-dominated regime in which the spring force is the only mechanism that keeps the follower in contact with the cam [1, 9]. Five interacting failure modes emerge in this regime, and they are not independent: float can trigger an impact that exceeds the residual coil-bind clearance, and resonance can amplify both.

- *Valve float* — the spring force can no longer overcome the inertial force on the cam deceleration ramp, the follower separates from the cam, and re-contact occurs in an uncontrolled manner [1, 9].
- *Valve bounce* — impact-driven oscillation at seating that degrades sealing and accelerates seat wear when the closing velocity grows with cam acceleration [1].
- *Coil bind* — adjacent coils make metal-to-metal contact, the spring stiffness rises abruptly, and the dynamic load is transmitted through the coil stack [10].
- *Spring fatigue* — of the order of 10^8 cycles per 200 h of operation under fluctuating shear loading between seat-preload and full-lift stress; any surface or shot-peening defect can dominate the failure process [10].
- *Resonance with cam harmonics* — coincidence of a cam harmonic with the spring natural frequency drives surge, accelerates coil-bind approach, and amplifies float [1].

The BowTie methodology, introduced next, provides a unified language to describe how each of these modes contributes to a single top event.

2.2. The BowTie methodology

The BowTie methodology organises a risk analysis around a single top event: threats are arranged on the left side connected to the top event through preventive barriers, and consequences are arranged on the right side connected through mitigative barriers [11]. Compared with FMEA, three advantages are relevant here: (i) the causal chain is made explicit rather than compressed into a severity–occurrence–detectability triplet; (ii) each preventive barrier can be associated with a measurable indicator and a pass criterion, which is the basis of the quantitative coupling proposed in this paper; and (iii) the diagrammatic representation is accessible to designers, operators, and reliability engineers simultaneously. The classical BowTie is qualitative; quantitative extensions in the target journal of this manuscript include Bayesian-network mapping [12, 8], Petri-net prognostics [13], adaptive bow-tie formulations [14], and cloud-model and Dempster–Shafer evidential extensions [15, 16]. The approach adopted here is complementary: each preventive barrier is mapped to a deterministic indicator from the analytical sizing of the component with a literature-anchored pass criterion, and the indicators are subsequently propagated through a Monte Carlo simulation to convert the nominal-point status into manufacturing-realistic probabilities of barrier compliance (Section 4.4).

2.3. Related work

Several authors have addressed valve-train and engine-component reliability from complementary perspectives. The work of Vélez Godiño et al. [4] on overhead valve trains in urban buses is a representative case study published in *Engineering Failure Analysis*: the authors use field data and metallurgical inspection to attribute the recurrent failure of an in-service valve train to a combination of operational misuse and inadequate inspection intervals, but they stop short of organising the findings around a barrier-based diagram. Soffritti et al. [5] report a four-cylinder diesel engine in which excessive wear at the rocker-arm/pushrod and rocker-arm/valve interfaces appeared after only 1 000 hours of operation; their root-cause analysis points to valve misalignment with respect to the seat insert as the dominant trigger. Both studies confirm that valve-train failures, once initiated, propagate quickly into other components and motivate the need for a preventive analysis framework.

Two further studies in *Engineering Failure Analysis* address elements of the valve-train system that are central to the present work. Xing et al. [6] investigate environment-assisted cracking in high-strength valve springs and show that the fatigue limit adopted at the design stage must be discounted to account for surface condition and lubricant chemistry; this finding directly supports the fatigue factor of safety adopted in Section 3.4. Zhao et al. [7] report lubrication failure in the camshaft bearings of a V6 diesel engine, identifying an asperity-contact regime that violates the design assumption of full hydrodynamic lubrication and reduces the effective service life of the bearing pair; their result reinforces the importance of the lubrication-related preventive barriers shown on the left side of the BowTie diagram of Section 3.3. Ko et al. [17] extend the spring-fatigue discussion beyond *Engineering Failure Analysis*, showing in *Scientific Reports* that surface flaws as small as a few micrometres on the oil-tempered wire of an automotive valve spring can reduce the fatigue life by an order of magnitude; this result motivates the conservative factor of safety adopted in Table 3.

Table 1 consolidates the four *Engineering Failure Analysis* case studies above and maps their reported root causes to the preventive (P) and mitigative (M) barriers and the deterministic indicators of the BowTie procedure constructed in Section 3. The mapping is direct for two of the four cases (Xing on environment-assisted cracking, caught by the Goodman fatigue indicator; and the operational/ maintenance signature of Vélez Godiño, caught by the mitigative side of the BowTie); is partial for the assembly-driven misalignment of Soffritti, which the qualitative bow flags but which lies outside the deterministic-indicator set adopted here; and is explicitly *outside* the present scope for the camshaft-bearing lubrication failure of Zhao, which would require an additional preventive barrier on the lubrication regime. The mixed mapping is itself an outcome of the comparison: it identifies which classes of failure the procedure already covers and which classes call for an extension of the barrier set, and it is examined in detail in Section 4.7.

The methodological literature on the BowTie diagram itself has matured in parallel. The review by de Ruijter and Guldenmund [11] remains the canonical reference for the qualitative formulation and catalogues the variations adopted in the chemical, aerospace, and nuclear industries. Khakzad et al. [8] propose a mapping of the BowTie into a Bayesian network so that the static qualitative diagram becomes dynamic and probabilistic; their construction is adopted in part by the BowTie tool used in the offshore industry. Aust and Pons [18] apply the methodology to aircraft engine borescope inspection and show that, even in the absence of probabilistic data, the BowTie diagram is an effective communication artefact between maintenance planners, inspectors, and reliability engineers. The methodology continues to be extended in *Reliability Engineering & System Safety* along two complementary fronts: Xie et al. [15] combine BowTie analysis with cloud-model risk-matrix methods to evaluate fire and explosion risks in oil depots where severity rubrics are inherently linguistic, and Sezer et al. [16] couple the BowTie diagram with Dempster–Shafer evidence theory and the HEART human-reliability framework for cargo-tank crack risk analysis. These extensions demonstrate the versatility of the underlying diagram when calibrated with domain-specific quantitative or evidential layers.

It is also relevant to acknowledge a parallel branch of the literature that extends the conventional FMEA in different directions: fuzzy-FMEA formu-

Table 1: Mapping of four documented valve-train and engine-component failure case studies from *Engineering Failure Analysis* to the preventive (P) and mitigative (M) barriers and the deterministic indicators of the BowTie procedure proposed in this work. The notation P1–P5, M1–M5 refers to the barriers introduced in Section 3.3, and the indicators are those of Table 3. Reported root causes are paraphrased from the cited papers; no values are reproduced.

Case study (component / engine)	Reported root cause	BowTie barrier triggered	Indicator that would flag it
Vélez Godiño et al. [4]; overhead valve train, urban bus	Operational misuse and inadequate inspection intervals	M3 (operating envelope at design) and M5 (predictive maintenance via vibration)	Mitigative side; outside the deterministic indicator set
Soffritti et al. [5]; rocker-arm / pushrod interfaces, four-cylinder diesel (∼1 000 h)	Valve misalignment relative to the seat insert	Threat: out-of-tolerance assembly → candidate geometric/quality barrier P6 (extension)	Outside the present indicator set; flagged qualitatively by the bow chain
Xing et al. [6]; high-strength valve springs (premature fatigue)	Environment-assisted cracking due to surface condition and lubricant chemistry	P4 (high-fatigue-resistance spring material)	Goodman factor of safety n_f with discounted shear endurance limit S_{se}
Zhao et al. [7]; camshaft bearings, V6 diesel	Asperity-contact regime violating the full-hydrodynamic-lubrication assumption	Threat: lubrication degradation; *outside the present preventive set*	Outside scope; motivates a lubrication-regime barrier in procedure extensions

lations and analytic-hierarchy-process-based weighting (AHP-FMEA) have been used in mechanical reliability to mitigate the loss of resolution of the multiplicative Risk Priority Number (RPN). The present work is complementary to those extensions: it does not aim to refine the FMEA score but to replace the single-score logic with a barrier-by-barrier deterministic check, and is therefore positioned closer to the BowTie tradition than to fuzzy/AHP-FMEA.

To the best of the authors' knowledge, no published study applies the BowTie methodology to the valve train of a high-rpm spark-ignition cylinder head, and no study couples the BowTie diagram to the analytical sizing of the intake valve and the valve spring as a reproducible barrier-evaluation procedure. The present work fills both gaps.

3. Methodology

3.1. System under study

The system under study is the valve train of a high-performance cylinder head conceived for the Volkswagen AP 1.8 platform. The base engine has a bore of 81.0 mm, a stroke of 86.4 mm, a total displacement of 1 781 cm^3, and a four-cylinder in-line architecture [3, § 1.1]. The cylinder head is of the pent-roof type, with inclined valves, asymmetric quench, gasoline direct injection (GDI), and an iridium spark plug; these features establish the mechanical and thermal context of the valve train but are not the subject of the risk analysis presented here. The reader is referred to Porto [3] for the full description of the cylinder head and the air-flow optimisation.

3.2. Mechanical and operational premises

The premises adopted for the present risk analysis are summarised in Table 2 and follow the design choices of the dissertation [3] and the associated sizing notes for the intake valve and the valve springs.

3.3. Construction of the qualitative BowTie

The BowTie diagram is constructed around the top event *loss of follower contact between valve and cam.* This formulation intentionally aggregates the three failure modes of Section 2.1 that share a single mechanical signature: valve float, valve bounce, and approach to coil bind. The remaining modes (resonance and fatigue) act as enabling threats rather than as the top event itself.

The threats placed on the left side of the bow are derived from the classical literature [1, 9, 19] and from the sizing constraints of the AP 1.8 head. They are: (i) insufficient spring stiffness for the inertial regime; (ii) excessive

Table 2: Mechanical and operational premises adopted for the risk analysis of the AP 1.8 valve train. Values taken from the dissertation [3] and the associated sizing notes for the intake valve and the valve springs.

Engine and operating regime		*Dual valve-spring assembly (Cr–Si steel, ASTM A877/A877M)*	
Bore	81.0 mm	Outer spring stiffness k_{out}	38.06 $\mathrm{N\,mm^{-1}}$
Stroke	86.4 mm	Inner spring stiffness k_{in}	61.43 $\mathrm{N\,mm^{-1}}$
Total displacement	1 781 $\mathrm{cm^3}$	Combined stiffness k_{total}	99.49 $\mathrm{N\,mm^{-1}}$
Cylinders	4 in line	Seat preload F_{seat}	780 N
Maximum engine speed	10 000 rpm	Force at full lift F_{open}	1 985.8 N
Camshaft fund. freq. $f_{\mathrm{cam,max}}$	83.3 Hz	Outer spring solid height	33.6 mm
Compression ratio	14.5 : 1	Inner spring solid height	32.0 mm
Valves, rocker and cam		Outer spring height at full lift	34.38 mm
Intake valve material	Ti–6Al–4V	Total spring assembly mass	152.72 g
Exhaust valve material	INCONEL 751	Effective spring mass ($m_{\mathrm{molas}}/3$)	50.91 g
Roller rocker / retainer / keeper	7075–T6 aluminium		
Camshaft material	SAE 8620 case-hardened		
Maximum valve lift	12.12 mm	*Direct translational mass (intake side)*	
Cam accel. peak (positive)	14 544 $\mathrm{m\,s^{-2}}$	Valve + retainer + keeper	43.48 g
Cam decel. peak (critical)	20 362 $\mathrm{m\,s^{-2}}$	With reflected rocker	49.59 g
		Direct translational mass (exhaust side)	
		Valve + retainer + keeper	54.25 g
		With reflected rocker	60.36 g

moving mass; (iii) aggressive cam profile; (iv) spring heating in operation; (v) out-of-tolerance thermal clearance; and (vi) resonance between cam harmonics and spring natural frequency. The preventive barriers placed between threats and the top event are: (i) dynamic sizing of the spring relative to the cam harmonics; (ii) dual or nested springs; (iii) low-mass tappets; (iv) high-fatigue-resistance spring material; and (v) modal analysis and dynamometer validation.

The consequences placed on the right side of the bow are: (i) valve–piston collision; (ii) catastrophic cylinder-head failure; (iii) immediate loss of compression; (iv) oil contamination by debris; and (v) unscheduled engine shutdown. The mitigative barriers between the top event and these consequences are: (i) electronic rev limiter; (ii) cam-position sensor; (iii) operating envelope defined at design stage; (iv) torque or fuel cut-off; and (v) predictive maintenance via vibration signals.

3.4. Quantitative coupling layer

To overcome the qualitative limitation of the classical BowTie, each of the five preventive barriers is associated with a deterministic indicator and a pass criterion, summarised in Table 3. The indicators are computed from the analytical sizing of the intake valve and of the valve springs, which is reported in the underlying dissertation [3] and in two associated sizing notes on the intake valve and the spring.

The dynamic-ratio criterion $R \geq 1.20$ corresponds to the project specification adopted in the AP 1.8 sizing and is consistent with the design practice

Table 3: Quantitative indicators associated with the preventive barriers of the BowTie diagram. The pass criteria follow conventional design rules in the internal-combustion-engine literature and the project specification of the AP 1.8 sizing.

Barrier	Indicator	Pass criterion
Dynamic follower contact	$R = F_{\text{spring,crit}}/F_{\text{inertia,crit}}$	$R \geq 1.20$ (project)
Spring natural frequency	$f_n = \frac{1}{2}\sqrt{k/m_s}$	$f_n/f_{\text{cam,max}} \geq 4$ [1]
Coil-bind margin (outer)	$M = H_{\text{open}} - H_{\text{cb}}$	$M \geq 1.5$ mm at 10 000 rpm
Fatigue (Goodman)	$1/n_f = \tau_a/S_{se} + \tau_m/S_{su}$	$n_f \geq 1$ [10]
Hot clearance†	$c(T_{\max})$	$c(T_{\max}) \geq 0$ over the full map

†Hot clearance is treated in this paper as a *qualitatively verified* barrier outside the deterministic layer; only a first-order order-of-magnitude estimate is provided in Section 4.2. A complete differential thermal model is identified as a priority direction for future work in Section 5.3.

for high-speed valve trains [1, 9, 19].

The natural-frequency criterion follows the rule of thumb that the first surge mode of the spring should sit above the dominant cam harmonics [1]. Two equivalent formulations are available in the literature: the one-degree-of-freedom approximation $f_n = \frac{1}{2}\sqrt{k/m_s}$ for a coil spring with both ends fixed [10, § 10-26], and the continuous-medium expression $f_n = (d/(2\pi N D_m^2))\sqrt{G/(2\rho)}$ [1, § 6.6], which reduce algebraically to the same result when m_s in the first form is taken as the active-coil mass. In the present analysis the conservative form $f_n = \frac{1}{2}\sqrt{k/m_{s,\text{total}}}$ is adopted with the *total* spring mass (including end coils), which yields the lower bound of the surge frequency and is the more demanding test for the rule-of-thumb threshold.

The coil-bind margin of 1.5–2.0 mm is the practical specification adopted in the AP 1.8 sizing to absorb manufacturing tolerances, mounting variations, and dynamic excitations typical of operation at 10 000 rpm. The fatigue criterion uses the Goodman alternating-mean factor of safety (Eq. 5) with the shear endurance limit and the ultimate shear strength of ASTM A877/A877M chrome–silicon valve-spring steel after shot peening and preset [10, 6, 17]. This paper does not propose new criteria; it organises them around the BowTie diagram.

3.5. Cross-check procedure

For the AP 1.8 head, the values of all indicators in Table 3 are computed under the operating envelope of Table 2. A barrier is declared *active* if the indicator satisfies the pass criterion; otherwise it is declared *at risk*. The single barrier with the smallest absolute margin is declared the *critical barrier*: it is the first that would fail under a parametric drift of the inputs (mass increase, stiffness loss through heating, harder cam profile, etc.). A tornado plot of the sensitivity of each indicator to its dominant input is reported in Section 4 (Fig. 2) to make the identification of the critical barrier transparent.

The qualitative BowTie of Section 3.3 and the quantitative layer of Section 3.4 are then cross-checked against a conventional FMEA/RPN ranking of the same threats. The comparison is discussed in Section 5.2.

4. Results

4.1. Populated BowTie diagram

Figure 1 shows the populated BowTie diagram for the AP 1.8 valve train at 10 000 rpm, around the top event *loss of follower contact between valve and cam*. The threats, preventive barriers, consequences, and mitigative barriers are those of Section 3.3.

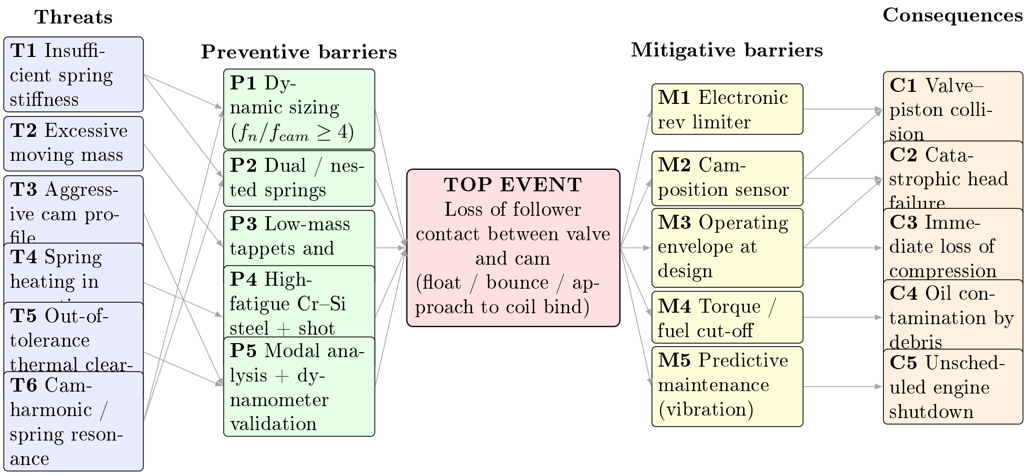

Figure 1: BowTie diagram of the valve train of the AP 1.8 cylinder head operating at 10 000 rpm. Threats (T1–T6) and preventive barriers (P1–P5) are arranged on the left side; consequences (C1–C5) and mitigative barriers (M1–M5) are arranged on the right side. The top event is the loss of follower contact between the valve and the cam, which encompasses valve float, valve bounce, and approach to coil bind.

4.2. Quantitative evaluation of the preventive barriers

The five quantitative indicators of Table 3 were evaluated for the AP 1.8 head under the operating envelope of Table 2. The numerical results are reported in Table 4 and discussed below.

Dynamic follower contact.. At the critical deceleration point of the cam ($a_{\max,-} = 20\,362\ \mathrm{m\,s^{-2}}$, $L_- = 8.747$ mm), the spring force is

$$F_{\text{spring},-} = F_{\text{seat}} + k_{\text{total}}\, L_- = 780 + 99.49 \times 8.747 = 1\,650.2\ \text{N}, \qquad (1)$$

and the inertial force on the intake side, computed with the direct translational mass of 43.48 g, is $F_{i,d,-,\text{adm}} = 885.3$ N, giving a dynamic ratio of $R_{d,-,\text{adm}} = 1.864$. With the equivalent mass that includes the reflected rocker (49.59 g), the dynamic ratio decreases to $R_{\text{eq},-,\text{adm}} = 1.634$. On the exhaust side, the larger mass (54.25 g translational direct) yields $R_{d,-,\text{esc}} = 1.494$ and $R_{\text{eq},-,\text{esc}} = 1.343$. All four values are above the $R \geq 1.20$ project criterion, so the follower-contact barrier is satisfied with a margin of at least 12 % (in the worst case, the exhaust side with reflected rocker).

Spring natural frequency.. The fundamental surge mode of a helical compression spring with both ends fixed is the first axial standing wave on the helical wire. For a spring of stiffness k and active mass m_s, the wave equation yields a fundamental frequency $\omega_1 = \pi\sqrt{k/m_s}\ \mathrm{rad\,s^{-1}}$, hence $f_n = \omega_1/(2\pi) = \frac{1}{2}\sqrt{k/m_s}$ Hz, which is the form given in Budynas and Nisbett [10, Eq. 10-26]. This expression is algebraically equivalent to the continuous-medium derivation $f_n = (d/(2\pi N D_m^2))\sqrt{G/(2\rho)}$ of Heywood [1, § 6.6]: substituting $k = Gd^4/(8D_m^3 N)$ and the active-coil wire mass $m_s = (\pi^2/4)\, N D_m d^2 \rho$ in $f_n = \frac{1}{2}\sqrt{k/m_s}$ recovers Heywood's expression exactly. In the present analysis the conservative lower bound is adopted by using the *total* spring mass (94.34 g for the outer and 47.11 g for the inner spring), which includes the inactive end turns and yields the lower bound of the surge frequency:

$$f_{n,\text{out}} = \tfrac{1}{2}\sqrt{38\,060\ \mathrm{N\,m^{-1}}/0.09434\ \mathrm{kg}} \approx 318\ \text{Hz}, \qquad (2)$$

$$f_{n,\text{in}} = \tfrac{1}{2}\sqrt{61\,430\ \mathrm{N\,m^{-1}}/0.04711\ \mathrm{kg}} \approx 571\ \text{Hz}. \qquad (3)$$

For comparison, replacing $m_{s,\text{total}}$ with the conventional effective mass $m_s/3$ (Rayleigh's quotient for a uniform coil spring with one end fixed) would yield $f_{n,\text{out}} \approx 551$ Hz and $f_{n,\text{in}} \approx 988$ Hz. Adopting the total spring mass therefore penalises the rule-of-thumb test against the cam harmonics by a factor of $\sqrt{3} \approx 1.73$ and is treated here as an explicit robustness check rather than a tolerance estimate. The camshaft fundamental frequency at 10 000 rpm engine speed ($f_{\text{cam,max}} = 10\,000/(2 \times 60) = 83.3$ Hz, the factor two accounting for the 4-stroke cam-to-engine speed ratio) gives the natural-frequency margins $f_{n,\text{out}}/f_{\text{cam,max}} \approx 3.81$ and $f_{n,\text{in}}/f_{\text{cam,max}} \approx 6.85$. The outer spring sits just below the conventional $f_n/f_{\text{cam}} \geq 4$ threshold [1] and is therefore at the limit of the preventive barrier; the inner spring is comfortably above the threshold against the cam fundamental. However, real cam profiles contain harmonic content up to the sixth to eighth harmonic, and the proximity of $f_{n,\mathrm{in}} \approx 571$ Hz to the sixth harmonic ($6\,f_{\mathrm{cam,max}} = 500$ Hz; about 14 % separation) is discussed in Section 4.5 and noted as a residual concern.

Coil-bind margin.. At the maximum valve lift of 12.12 mm, the outer spring is compressed to a height of $H_{\mathrm{open}} = 34.38$ mm. With a solid height of $H_{\mathrm{cb,out}} \approx 33.6$ mm (from $N_{t,\mathrm{out}}\, d_{\mathrm{out}}$), the residual margin to coil bind is

$$M_{\mathrm{out}} = H_{\mathrm{open}} - H_{\mathrm{cb,out}} = 34.38 - 33.6 = 0.78 \text{ mm}, \tag{4}$$

which is below the project specification of $M \geq 1.5$ mm at 10 000 rpm. In the concentric dual-spring assembly the inner and outer springs share the same retainer and therefore undergo the same axial compression at full lift, so H_{open} is identical for both. The inner spring, with $H_{\mathrm{cb,in}} \approx 32.0$ mm, therefore yields $M_{\mathrm{in}} = H_{\mathrm{open}} - H_{\mathrm{cb,in}} = 34.38 - 32.0 = 2.38$ mm, comfortably above the criterion. The outer-spring coil-bind margin is therefore the only barrier that does not satisfy its pass criterion under the current configuration of the AP 1.8 head.

Fatigue stress.. The peak shear stress at full lift, computed with the Wahl-corrected formula $\tau_{\max} = K_w\, 8FD_m/(\pi d^3)$, is $\tau_{\mathrm{out}} \approx 655$ MPa for the outer spring ($K_{w,\mathrm{out}} = 1.239$, $D_{m,\mathrm{out}} = 30.2$ mm, $d_{\mathrm{out}} = 4.8$ mm) and $\tau_{\mathrm{in}} \approx 1\,232$ MPa for the inner spring ($K_{w,\mathrm{in}} = 1.329$, $D_{m,\mathrm{in}} = 19.0$ mm, $d_{\mathrm{in}} = 4.0$ mm). Because the spring operates between a non-zero seat preload and the full-lift load, the proper criterion is the alternating-mean Goodman criterion [10, § 10-7] rather than a single-point comparison with an allowable. With the corresponding seat-load shear stresses $\tau_{\mathrm{out,seat}} \approx 257$ MPa and $\tau_{\mathrm{in,seat}} \approx 484$ MPa, the alternating and mean components are $\tau_{a,\mathrm{out}} = 199$ MPa, $\tau_{m,\mathrm{out}} = 456$ MPa and $\tau_{a,\mathrm{in}} = 374$ MPa, $\tau_{m,\mathrm{in}} = 858$ MPa.

For shot-peened ASTM A877/A877M chrome–silicon valve-spring steel, typical strengths are an ultimate tensile strength $S_{ut} \approx 1\,900$ MPa, an ultimate shear strength $S_{su} \approx 0.67\, S_{ut} \approx 1\,270$ MPa, and a shear endurance limit after shot peening and surface control of $S_{se} \approx 465$ MPa [10, § 10-30]. The Goodman factor of safety is then

$$\frac{1}{n_f} = \frac{\tau_a}{S_{se}} + \frac{\tau_m}{S_{su}}. \tag{5}$$

For the outer spring this gives $1/n_{f,\mathrm{out}} = 199/465 + 456/1\,270 \approx 0.428 + 0.359 = 0.787$, hence $n_{f,\mathrm{out}} \approx 1.27$ (active, with positive but not generous margin). For the inner spring, $1/n_{f,\mathrm{in}} = 374/465 + 858/1\,270 \approx 0.804 + 0.676 = 1.480$, hence $n_{f,\mathrm{in}} \approx 0.68$, which is below unity and indicates that the baseline configuration of the inner spring does not satisfy the Goodman criterion. The inner-spring fatigue barrier is therefore reclassified as **at risk** in Table 4, alongside the outer-spring coil-bind margin. This finding is consistent with Xing et al. [6], Ko et al. [17], who report that small surface flaws and environment-assisted cracking substantially reduce the fatigue life of high-strength valve springs loaded close to their nominal allowable.

The shear endurance limit $S_{se} \approx 465$ MPa quoted above is the ambient-temperature value catalogued for shot-peened ASTM A877 chrome–silicon wire. Under sustained operation at 10 000 rpm the valve spring sees a temperature elevation associated with hysteretic self-heating and conductive exposure to the oil-splash environment; the steady-state spring-coil temperature for high-rpm SI valve springs is documented in the range 100–130 °C [19]. The Marin-type temperature factor of Budynas and Nisbett [10, Eq. 7-19] for ground steel surfaces, evaluated at $T = 130$ °C (≈ 266 °F), yields a temperature correction factor $k_d \approx 1.02$, i.e. no derating relative to ambient over the expected operating range. A conservative counter-bound, applying a uniform 5 % temperature penalty ($S_{se,T} = 442$ MPa), gives $n_{f,\text{out}} \approx 1.24$ and $n_{f,\text{in}} \approx 0.66$: both indicator classifications (outer active, inner at risk) are preserved, and the identification of the inner-spring Goodman deficit is robust to the temperature treatment within ± 5 % on S_{se}. The barrier status of Table 4 is therefore not contingent on the ambient-versus-operating temperature distinction within the documented operating envelope; an out-of-envelope spring temperature in the 180–220 °C range (e.g. degraded oil-splash cooling) would invalidate this conclusion and is identified in Section 5.3 as an off-design follow-on.

Hot clearance.. A complete coupled thermal–structural finite-element analysis of the cylinder head is outside the scope of the present study, but a *differential* first-order estimate of the magnitude can be obtained in closed form by combining the linear thermal expansion of the steel valve stem and the aluminium head along the load-bearing path that closes the rocker–valve-tip clearance. Under hot operation the exhaust valve stem (length $L_h = 180$ mm, INCONEL 751 with $\alpha_{\text{Fe}} \approx 13 \times 10^{-6}$ K^{-1} [20]) elongates by

$$\Delta L_{\text{stem}} = \alpha_{\text{Fe}}\, L_h\, \Delta T_{\text{stem}} \approx 13 \times 10^{-6} \times 180 \times 600 \approx 1.40 \text{ mm}, \qquad (6)$$

for a stem temperature rise of $\Delta T_{\text{stem}} \approx 600$ K above cold-assembly (consistent with the $\sim$650 °C peak exhaust-valve-stem temperature reported for high-output SI engines [19]). The aluminium head between the valve seat and the rocker pivot, of effective length $L_{\text{Al}} \approx 100$ mm (typical OHC architecture [19]) and $\alpha_{\text{Al}} \approx 23 \times 10^{-6}$ K^{-1}, expands in the *opposite* sense (the pivot retreats from the valve tip, opening the clearance). For a head temperature rise of $\Delta T_{\text{head}} \approx 150$ K (steady-state operating range for the aluminium structure, in line with documented head temperatures of 130–180 °C above ambient [1]), the head contribution is

$$\Delta L_{\text{head}} = \alpha_{\text{Al}}\, L_{\text{Al}}\, \Delta T_{\text{head}} \approx 23 \times 10^{-6} \times 100 \times 150 \approx 0.35 \text{ mm}. \tag{7}$$

The differential closure of the clearance is therefore

$$\Delta c_{\text{diff}} = \Delta L_{\text{stem}} - \Delta L_{\text{head}} \approx 1.40 - 0.35 = 1.05 \text{ mm}, \tag{8}$$

which is materially less restrictive than the stem-only upper bound of 1.40 mm. The cold mechanical clearance must satisfy $c_{\text{cold}} \geq \Delta c_{\text{diff}} + c_{\text{hot,min}}$ with $c_{\text{hot,min}} \approx 0.20$ mm representing the minimum positive operating clearance required for reliable valve seating [19], yielding a pass criterion of $c_{\text{cold}} \geq 1.25$ mm. This deterministic indicator is added to Table 4 as $c_{\text{cold,req}} = 1.25$ mm, replacing the qualitative "verified" status of the earlier formulation. The estimate is bracketed by two limiting cases: the no-compensation bound ($\Delta L_{\text{head}} = 0$, $c_{\text{cold,req}} = 1.60$ mm) and a maximally compensating bound ($\Delta T_{\text{head}} = 200$ K, $c_{\text{cold,req}} = 1.13$ mm). The bracket $c_{\text{cold,req}} \in [1.13, 1.60]$ mm is compatible with the typical cold-clearance specification of 1.5 mm for high-rpm performance heads quoted in Heisler [19], so the barrier is classified as *active* in the baseline configuration. A coupled thermal–structural finite-element analysis would refine the head-side path length L_{Al} and the localised ΔT_{head} field and is identified in Section 5.3 as a priority follow-on study.

Table 4: Quantitative results for the five preventive barriers of the BowTie diagram applied to the AP 1.8 valve train at 10 000 rpm. Values traced to the sizing of the intake valve and the valve springs.

Barrier	Indicator	Value	Status
Dynamic follower contact (intake, direct mass)	$R_{d,-,\text{adm}}$	1.864	active (≥ 1.20)
Dynamic follower contact (intake, with rocker)	$R_{\text{eq},-,\text{adm}}$	1.634	active (≥ 1.20)
Dynamic follower contact (exhaust, direct mass)	$R_{d,-,\text{esc}}$	1.494	active (≥ 1.20)
Dynamic follower contact (exhaust, with rocker)	$R_{\text{eq},-,\text{esc}}$	1.343	active (≥ 1.20)
Natural frequency (outer spring)	$f_{n,\text{out}}/f_{\text{cam,max}}$	3.81	at limit (< 4)
Natural frequency (inner spring)	$f_{n,\text{in}}/f_{\text{cam,max}}$	6.85	active (≥ 4)
Coil-bind margin (outer spring)	M_{out}	0.78 mm	**at risk** (< 1.5 mm)
Coil-bind margin (inner spring)	M_{in}	2.38 mm	active (≥ 1.5 mm)
Fatigue Goodman FoS (outer spring)	$n_{f,\text{out}}$	1.27	active (≥ 1)
Fatigue Goodman FoS (inner spring)	$n_{f,\text{in}}$	0.68	**at risk** (< 1)
Hot clearance (differential, exhaust)	$c_{\text{cold,req}}$	1.25 mm (bracket [1.13, 1.60])	active ($c_{\text{cold}} \geq c_{\text{cold,req}}$)

4.3. Critical-barrier identification

A tornado plot of the sensitivity of each indicator to its dominant input is shown in Fig. 2. For each barrier, the input perturbed by ±10 % is the dominant parameter governing the indicator: the spring stiffness k for the natural-frequency margin, the maximum cam lift $L_{\mathrm{cam}}^{\max}$ for the coil-bind margin, the moving mass m for the dynamic-ratio margin, and the shear-stress amplitude τ_a for the Goodman fatigue margin. The $\pm 10\,\%$ perturbation is adopted as a uniform stress test rather than a tolerance estimate, providing an upper-bound sensitivity envelope for the worst-case parameter drift. The actual ASTM A877/A877M specification tolerances are tighter and unequal by parameter (typically $\pm 1.5\,\%$ on free length, $\pm 0.5\,\%$ on wire diameter, and $\pm 3\,\%$ on shear modulus); a refined sensitivity analysis based on those non-uniform tolerances is identified as future work. The bar length corresponds to the absolute change in the residual margin under that perturbation. The analysis identifies **three** barriers as exposed to parametric drift: the *outer-spring coil-bind margin* ($M_{\mathrm{out}} = 0.78$ mm, violating its pass criterion already in the baseline configuration), the *inner-spring fatigue Goodman factor of safety* ($n_{f,\mathrm{in}} = 0.68$, below unity in the baseline configuration), and the *outer-spring natural-frequency margin* ($f_{n,\mathrm{out}}/f_{\mathrm{cam,max}} = 3.81$, just below the conventional $4\times$ rule of thumb).

The findings indicate that the AP 1.8 dual-spring assembly has *two* concurrent at-risk barriers in the baseline configuration: the outer-spring coil-bind margin and the inner-spring fatigue factor of safety. They are not redundant: they affect different springs (outer vs inner) and act through different physical mechanisms (geometric clearance vs cyclic shear loading). The outer-spring natural-frequency margin sits just below the conventional threshold and is treated as a residual concern. The sizing note for the springs already prescribes the geometric correction for the outer-spring coil-bind — raising the installed height to $H_{\mathrm{inst,ideal}} = 47.72$ mm and the free length to $L_{0,\mathrm{ideal}} = 55.56$ mm to recover an $M \approx 2.0$ mm margin. The inner-spring fatigue concern is not addressed by that geometric change and would require a different intervention: increasing the wire diameter d_{in}, reducing the spring index $C_{\mathrm{in}} = D_{m,\mathrm{in}}/d_{\mathrm{in}}$, or adopting a higher-grade chrome–silicon–vanadium steel with a larger shear endurance limit. The two findings together reinforce the value of the quantitative-coupled BowTie approach: a single-point comparison of $\tau_{\max}$ with an upper-bound allowable, in the spirit of conventional FMEA, would have classified the inner-spring fatigue as nominally acceptable, whereas the Goodman criterion exposes the deficit unambiguously.

4.4. Probabilistic refinement under ASTM A877/A877M tolerances

The $\pm 10\,\%$ sensitivity of Section 4.3 is a uniform stress test rather than a tolerance estimate. To convert it into a manufacturing-realistic probability of barrier compliance, the deterministic indicators were re-evaluated under the catalogued ASTM A877/A877M spring-wire tolerances and the standard

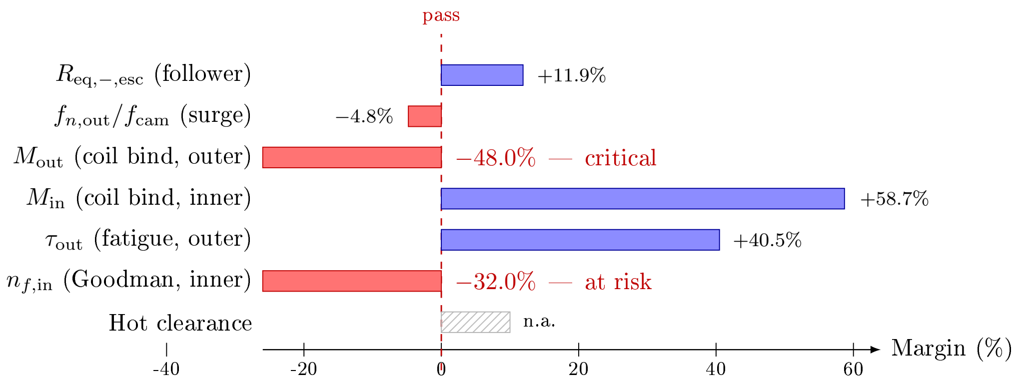

Figure 2: Tornado plot of the sensitivity of each preventive-barrier indicator to its dominant input under a $\pm 10\,\%$ perturbation. Two indicators (in red) violate their pass criteria already in the baseline configuration and are identified as critical preventive barriers of the AP 1.8 valve train at 10 000 rpm: the outer-spring coil-bind margin M_{out} (-48.0%) and the inner-spring fatigue Goodman factor of safety $n_{f,\mathrm{in}}$ (-32.0%). The outer-spring natural-frequency margin $f_{n,\mathrm{out}}/f_{\mathrm{cam,max}}$ (-4.8%) sits just below the conventional $4\times$ rule-of-thumb threshold and is treated separately as a residual concern. The uniform $\pm 10\,\%$ perturbation is conservative relative to the ASTM A877/A877M tolerance set used in the probabilistic refinement of Section 4.4.

strength dispersion of Budynas and Nisbett [10, § 10-30]. The propagation was performed by a Monte Carlo simulation with $N = 2 \times 10^5$ truncated-Gaussian draws of each input, with the three-sigma band set equal to the catalogued tolerance for each parameter:

Parameter	3σ band	Source
Free length L_0	$\pm 1.5\,\%$	ASTM A877/A877M
Wire diameter d	$\pm 0.5\,\%$	ASTM A877/A877M
Mean coil diameter D_m	$\pm 1.0\,\%$	ASTM A877/A877M
Active coils N_a	± 0.5 turn	ASTM A877/A877M
Shear modulus G	$\pm 3.0\,\%$	Budynas and Nisbett [10, Table 10-4]
Shear endurance S_{se}	$\pm 8.0\,\%$	Budynas and Nisbett [10, § 10-30]
Ultimate shear S_{su}	$\pm 4.0\,\%$	idem
Seat preload F_{seat}	$\pm 3.0\,\%$	assembly measurement
Open height H_{open}	$\pm 0.5\,\%$	cam-train assembly

The sensitivity coefficients are derived analytically from the Wahl stress formula $\tau = K_w\, 8FD_m/(\pi d^3)$, Shigley's spring stiffness $k = Gd^4/(8D_m^3 N_a)$, and Shigley's surge frequency $f_n = (d/(2\pi N_a D_m^2))\sqrt{G/(2\rho)}$. The script that reproduces the simulation (random seed 20260519, NumPy 2.2.6) is available as supplementary material.

The resulting per-trial distributions of the three indicators are summarised in Table 5. Three findings emerge:

1. *Inner-spring Goodman deficit is robust.* The Goodman factor of safety of the inner spring is below unity in **100 %** of the simulated tolerance bracket ($n_{f,\text{in}} \in [0.637, 0.716]$ between the 5 % and 95 % quantiles). The baseline finding is not an artefact of nominal-point evaluation: under the catalogued ASTM A877 manufacturing band, the inner spring fails the Goodman criterion with certainty. This strengthens, rather than challenges, the deterministic conclusion of Section 4.3.
2. *Outer-spring coil-bind margin is statistically critical.* Within the catalogued ASTM A877 tolerance band, the outer-spring coil-bind margin falls below the project criterion of $M \geq 1.5$ mm in **77.9 %** of the trials, and the 5 % quantile is $M_{\text{out}} = -0.76$ mm, i.e. *actual contact* with the bind condition occurs in approximately 5 % of the distribution. The barrier is therefore not merely under-margined: a non-trivial fraction of the manufactured assemblies under nominal tolerances would experience hard coil bind at maximum lift before geometric correction.
3. *New finding: marginal inner-spring coil-bind.* The nominal $M_{\text{in}} = 2.38$ mm appears to satisfy the criterion comfortably; under tolerance propagation, however, $P[M_{\text{in}} < 1.5 \text{ mm}] = 16.2\,\%$. The barrier is therefore not as safe as the deterministic value suggests, and an analogous geometric verification of the installed height is recommended for the inner spring as well. This is a finding that the deterministic analysis of Section 4.2 does not surface and is a direct value-added contribution of the probabilistic refinement.

The Monte Carlo refinement converts the qualitative "at-risk" status of the two critical barriers identified by the deterministic analysis into quantitative probabilities of compliance failure under manufacturing-realistic dispersion, and surfaces a third barrier (M_{in}) that the nominal-point analysis classifies as safe but the probabilistic refinement re-classifies as marginal. These probabilities map naturally onto the BowTie barrier-status language: $P[\text{fail}] > 50\,\%$ is reported as *at risk*, $10\,\% < P[\text{fail}] \leq 50\,\%$ as *marginal*, and $P[\text{fail}] \leq 10\,\%$ as *active*. With this convention, the AP 1.8 baseline configuration shows two at-risk barriers, one marginal barrier, and the surge-ratio outer-spring barrier on the boundary of the at-risk classification.

4.5. Lift profile and natural-frequency map

For completeness, Fig. 3 shows the position of the two critical points of the cam profile (positive-acceleration peak and deceleration peak) on the lift–versus–crank-angle plane; Fig. 4 compares the spring natural frequencies with

Table 5: Monte Carlo refinement of the preventive-barrier indicators under ASTM A877/A877M tolerances and the catalogued strength dispersion of Budynas and Nisbett [10, § 10-30]. The 5 % and 95 % quantiles span the central 90 % of the manufactured population. The probability of barrier compliance failure is reported with respect to the project pass criteria of Table 3.

Indicator	Median	5 % quantile	95 % quantile	P[barrier fail]
Outer spring				
Coil-bind margin M_{out} (mm)	0.779	−0.760	2.329	77.9 %
Goodman FoS $n_{f,\text{out}}$	1.270	1.197	1.346	< 0.01 %
Surge ratio $f_{n,\text{out}}/f_{\text{cam,max}}$	3.816	3.642	4.007	94.4 %
Inner spring				
Coil-bind margin M_{in} (mm)	2.379	0.911	3.840	16.2 %
Goodman FoS $n_{f,\text{in}}$	0.675	0.637	0.716	100.0 %
Surge ratio $f_{n,\text{in}}/f_{\text{cam,max}}$	6.853	6.538	7.195	< 0.01 %

the first cam harmonics across the rotational-speed range up to 10 000 rpm engine speed.

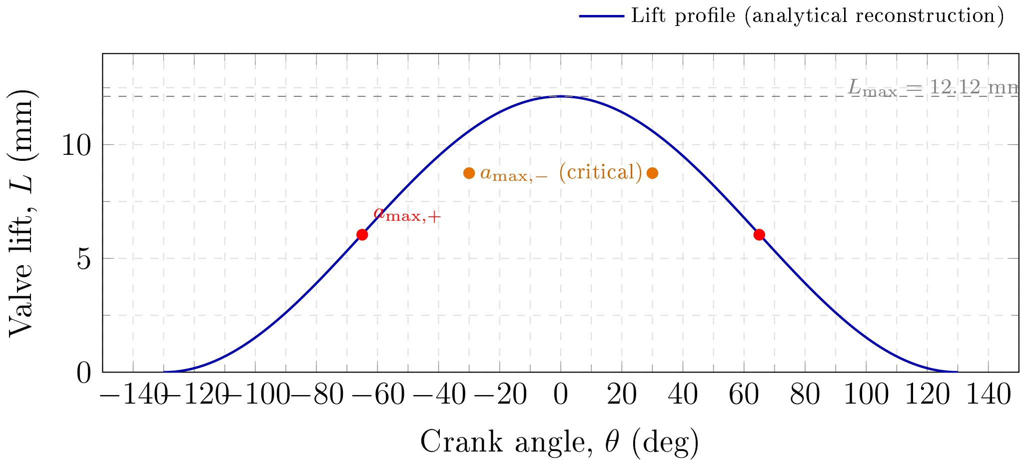

Figure 3: Schematic position of the two critical points of the cam profile of the AP 1.8 high-performance head: positive-acceleration peak ($a_{\text{max},+} = 14\,544$ m s^{-2} at $L_+ = 6.04$ mm) and deceleration peak ($a_{\text{max},-} = 20\,362$ m s^{-2} at $L_- = 8.747$ mm). The full lift profile is generated from the cam sizing of the intake valve.

4.6. Parametric envelope across engine speed

A single-design-instance analysis ($n = 1$) does not by itself demonstrate that the procedure adds value across the operational range; to address this, the dynamic-ratio indicator R and the surge-ratio indicator f_n/f_{cam} were swept across the engine-speed range 6 000–12 000 rpm at fixed AP 1.8 head

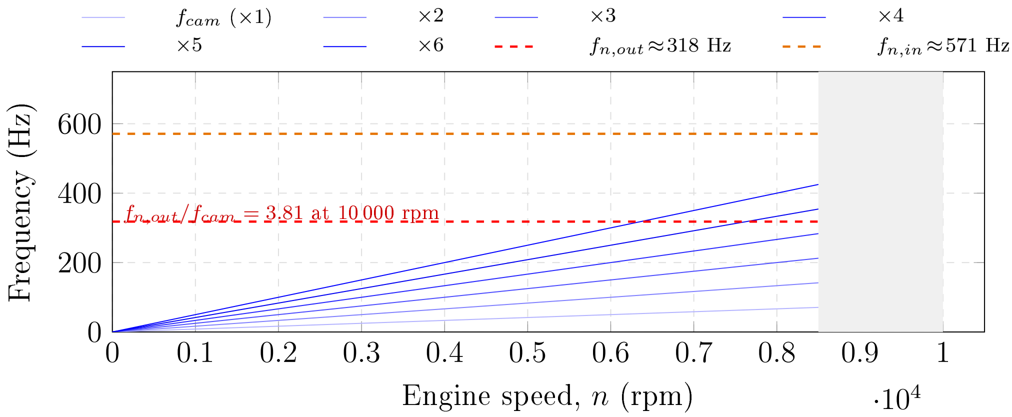

Figure 4: Spring natural frequencies $f_{n,\text{out}}$ and $f_{n,\text{in}}$ compared with the first six cam harmonics ($n = 1, \ldots, 6$) across the engine-speed range up to 10 000 rpm. The shaded region marks the operating regime; the dashed horizontal lines mark $f_{n,\text{out}} \approx 318$ Hz and $f_{n,\text{in}} \approx 571$ Hz.

geometry. The dynamic-ratio indicator scales analytically as $R(\omega) = R(\omega_0)\,(\omega_0/\omega)^2$ (cam acceleration $\propto \omega^2$ while spring force at the critical lift is ω-independent), and the surge-ratio scales as $f_n/f_{\text{cam}} \propto 1/\omega$. Table 6 reports the worst-case exhaust-side equivalent dynamic ratio $R_{\text{eq},-,\text{esc}}$ and the outer-spring surge ratio across the sweep.

Table 6: Parametric sweep of the two speed-sensitive indicators across the engine-speed range 6 000–12 000 rpm for the AP 1.8 head at fixed geometry. The dynamic-ratio criterion is $R \geq 1.20$ and the surge-ratio rule-of-thumb is $f_n/f_{\text{cam}} \geq 4$. The intersection of the two pass criteria defines an analytical safe-operation envelope of approximately 6 000–10 500 rpm for the baseline configuration; barriers transition to at-risk above this band and the loss-of-contact boundary ($R \to 1$) is crossed near 11 500 rpm.

Engine speed (rpm)	$R_{\text{eq},-,\text{esc}}$	$f_{n,\text{out}}/f_{\text{cam}}$	Envelope status
6 000	3.73	6.36	comfortably active
7 000	2.74	5.45	active
8 000	2.10	4.77	active
9 000	1.66	4.24	active
10 000	1.34	3.82	at limit (surge)
10 500	1.22	3.63	at limit (dynamic)
11 000	1.11	3.47	at risk
11 500	1.02	3.32	loss-of-contact boundary
12 000	0.93	3.18	beyond envelope

The parametric sweep produces three operational boundaries that the deterministic framework predicts in closed form: the surge-ratio rule-of-thumb is crossed at approximately 9 800 rpm; the dynamic-ratio project criterion ($R \geq 1.20$) is crossed at approximately 10 500 rpm; and the loss-of-contact boundary ($R = 1$) is crossed at approximately 11 500 rpm. The AP 1.8 design point of 10 000 rpm therefore sits in the most demanding region of the safe envelope, with negligible margin to the dynamic-ratio criterion and only $\approx 5\,\%$ geometric margin on the cam-profile critical-deceleration point. The sweep extends the single-point analysis of Section 4.2 into an operating-envelope view without re-running the analytical sizing, illustrating that the procedure is not bound to a single rpm and can be transposed to other operating points or other high-rpm cylinder heads by the same closed-form scaling rules.

4.7. Benchmark against documented failure cases

The deterministic indicators of Table 4 establish whether the AP 1.8 head satisfies its design margins; they do not by themselves demonstrate that the BowTie procedure would correctly flag the failure modes encountered in service by other engines. To address that gap without introducing experimental data that is not available for the AP 1.8 configuration, the four valve-train and engine-component failure case studies catalogued in Table 1 are now re-examined as a benchmark set: each reported root cause is matched with the BowTie barrier that, on the bow diagram of Section 4.1, would have been the controlling element, and the deterministic indicator (if any) that would have exposed the deficit at the design stage is identified.

Construct-validity disclosure. The four published case studies surveyed below were inputs to the formulation of the preventive barrier set P_1–P_5 of Section 3.3: they were read before the bow diagram was finalised, and the threat list was constructed to cover their physical mechanisms. The exercise that follows is therefore a *construct-validity check* (does the framework, as designed, account for known failure mechanisms?) rather than a *predictive-validity* test (would the framework predict unseen failures?). Predictive validity in the strict sense requires either application to an unrelated future failure or a test-rig campaign on the AP 1.8 head itself; both are identified as priority follow-on work in Section 6. Acknowledged construct-validity testing is nevertheless an established step in the development of risk-assessment procedures, and the outcome of the present check is informative even under that more limited claim.

Four outcomes are possible: *caught on the deterministic side* (a preventive-barrier indicator flags the deficit numerically), *caught on the mitigative side* (the failure mechanism is an operational signature addressed by a mitigative barrier and outside the preventive deterministic layer by construction), *flagged qualitatively only* (the threat is in the bow but no quantitative indicator exists in the present formulation), or *outside the present scope* (the failure mechanism falls outside the barrier set adopted here and would require an extension of the procedure). The four cases populate all four outcomes, and this distribution itself is informative: the BowTie procedure as formulated maps three of the four root causes onto its own preventive or mitigative layer (Xing on the deterministic side; Vélez Godiño on the mitigative side; Soffritti at the qualitative level), while the fourth case (Zhao) defines the boundary of the current barrier set and motivates a documented extension.

Xing et al. [6] — caught on the deterministic side.. The premature fatigue of high-strength valve springs reported by Xing is attributed to environment-assisted cracking, which depresses the effective shear endurance limit S_{se} below the catalogued allowable for shot-peened wire. The preventive barrier P4 (*high-fatigue-resistance spring material*) maps directly to the Goodman factor of safety n_f of Eq. (5), evaluated with a S_{se} discounted for surface condition and lubricant chemistry as recommended by the case study itself. In the present analysis the discounted $S_{se} \approx 465$ MPa for shot-peened ASTM A877/A877M (see Section 4.2, "Fatigue stress") is precisely the value advocated by Xing. The fact that the same indicator, applied to the AP 1.8 inner spring, returns $n_{f,\mathrm{in}} \approx 0.68 < 1$ is the strongest demonstration of the procedure: the indicator that diagnosed Xing's premature failure at a generic level exposes the inner-spring deficit of the AP 1.8 head before the prototype is built.

Vélez Godiño et al. [4] — caught on the mitigative side.. The recurrent in-service failure of the overhead valve train of urban buses is attributed by Vélez Godiño et al. to a combination of operational misuse and inadequate inspection intervals — a maintenance and operational signature, not a design-margin signature. By construction, this class of failure falls on the mitigative side of the bow rather than on the preventive side: barriers M3 (*operating envelope defined at design stage*) and M5 (*predictive maintenance via vibration signals*) are the controlling elements. The deterministic indicators of Table 3 would not have flagged this case at the design stage — and this is the correct behaviour: the case is, by its own description, an operational problem expressed at the maintenance interface. The BowTie diagram visualises this distinction explicitly, whereas a single-score RPN ranking blends preventive and mitigative dimensions into one number.

Soffritti et al. [5] — flagged qualitatively only.. The premature wear at the rocker-arm/pushrod and rocker-arm/valve interfaces in a four-cylinder diesel engine is attributed by Soffritti et al. to valve misalignment with respect to the seat insert. The threat "out-of-tolerance assembly geometry" is not explicitly listed among T_1–T_6 in the bow of Section 3.3, but it is a direct extension of the existing threat structure: it sits next to T_2 (*excessive moving mass*) and T_3 (*aggressive cam profile*) as a geometric departure from the sizing assumption. No deterministic indicator in Table 3 maps to it: the case is therefore *flagged qualitatively only* by the present procedure. This outcome motivates a candidate extension of the preventive barrier set — an assembly/tolerance barrier P_6 with a dimensionless stem-to-seat runout indicator ρ_{run} (specified in Section 6.2) — which is identified as a direction for future work.

Zhao et al. [7] — outside the present scope.. The lubrication failure of the camshaft bearings of a V6 diesel engine is attributed by Zhao et al. to an asperity-contact regime that violates the design assumption of full hydrodynamic lubrication. The present BowTie is centred on the spring–cam dynamic interaction and does not include a lubrication-regime preventive barrier; the case therefore lies outside the scope of the procedure as formulated here. The honest outcome of the comparison is that the procedure would have *missed* this failure mode — and that explicit recognition is itself the useful finding: an extension of the procedure with an additional preventive barrier on the lubrication regime, specified as the Stribeck-curve minimum film-thickness ratio $\Lambda_{\min}$ of Section 6.2 with pass criterion $\Lambda_{\min} \geq 1.5$, closes the gap before the framework can claim coverage of the full set of valve-train-related failure mechanisms.

Summary of the qualitative validation.. Of the four documented case studies surveyed in Table 1, one is caught quantitatively by the deterministic indicator set (Xing, via the Goodman factor of safety), one is caught on the mitigative side as designed (Vélez Godiño), one is flagged qualitatively but lies outside the deterministic indicator set (Soffritti, motivating a candidate P_6), and one is outside the present scope altogether (Zhao, motivating a lubrication-regime extension). The pattern is consistent: the procedure correctly distinguishes between design-margin failures (where the deterministic indicators are diagnostic), operational/maintenance failures (where the mitigative side is diagnostic), and failure mechanisms outside the present barrier set (where the comparison flags the limitation). This qualitative validation supports the use of the procedure as a design-stage check while confirming that an empirical campaign on a dedicated test rig — specified in Section 6 — remains necessary for quantitative validation of the indicator margins reported here.

5. Discussion

5.1. Implications for cylinder-head design

The quantitative-coupled BowTie of Section 4 provides a single artefact that links each design decision in the high-performance cylinder head to the risk it controls. The analysis identified two concurrent at-risk barriers under the baseline configuration of the AP 1.8 head: (i) the *outer-spring coil-bind margin* ($M_{\text{out}} = 0.78$ mm against a project criterion of $M \geq 1.5$ mm), and (ii) the *inner-spring fatigue Goodman factor of safety* ($n_{f,\text{in}} \approx 0.68$ against $n_f \geq 1$). The outer-spring natural-frequency margin ($f_{n,\text{out}}/f_{\text{cam,max}} \approx 3.81$) sits just below the conventional rule-of-thumb threshold of 4 and is treated as a residual concern. The remaining barriers — dynamic follower contact ($R \geq 1.343$ in the worst case, against the project criterion of $R \geq 1.20$), inner-spring coil-bind margin ($M_{\text{in}} = 2.38$ mm), and outer-spring fatigue ($n_{f,\text{out}} \approx 1.27$) — satisfy their pass criteria with non-uniform margins.

The fact that the two at-risk barriers affect *different* springs (outer for coil-bind, inner for fatigue) and act through *different* physical mechanisms (geometric clearance vs cyclic shear loading) is significant: a design iteration that addresses only one of the two findings will leave the other unresolved. The sizing note for the AP 1.8 springs already prescribes the geometric correction for the outer-spring coil-bind (raising the installed height to $H_{\text{inst,ideal}} = 47.72$ mm and the free length to $L_{0,\text{ideal}} = 55.56$ mm to achieve $M \approx 2.0$ mm). The inner-spring Goodman deficit, by contrast, calls for a metallurgical or geometric change of a different nature: either increasing the wire diameter d_{in} to lower the shear stress at a given load, reducing the spring index $C_{\text{in}} = D_{m,\text{in}}/d_{\text{in}}$, or specifying a higher-grade steel (e.g. chrome–silicon–vanadium ASTM A878) with a larger corrected shear endurance limit S_{se}. Concerning the residual outer-spring natural-frequency concern, a parallel correction would adjust either the active coil count $N_{a,\text{out}}$ or the spring index C_{out} to push $f_{n,\text{out}}$ further from the dominant cam harmonics. The BowTie diagram makes the three trade-offs simultaneously visible to the design team.

The remaining barriers retain non-zero margins and therefore satisfy the literature-anchored pass criteria, but Table 4 shows that the margins are not uniform. This non-uniformity is a useful input for design iteration: barriers with comfortable margins (e.g. the inner-spring coil-bind margin or the outer-spring fatigue stress) are candidates for relaxation if doing so frees mass or simplifies manufacturing; barriers close to the pass threshold should be the focus of dynamometer validation as soon as a physical prototype is available.

5.2. Comparison with traditional FMEA/RPN

Table 7 contrasts the present BowTie analysis with a conventional FMEA of the same six threats. Three observations follow.

1. *Granularity of the causal chain*: the BowTie diagram makes the linkage between threat, preventive barrier, top event, and consequence visible in a single artefact, whereas FMEA collapses the causal chain into the multiplicative RPN. For a high-rpm valve train, this granularity is necessary because several threats share the same preventive barriers (for example, both *insufficient stiffness* and *spring heating* are addressed by the spring-natural-frequency barrier).
2. *Quantitative anchoring*: in FMEA, severity, occurrence, and detectability are typically assigned on integer scales (1–10) by expert judgement. The quantitative-coupled BowTie replaces these ordinal scores with deterministic indicators that can be reproduced by any reader who has access to the sizing calculations.
3. *Communication*: the BowTie diagram is accessible to designers, operators, and reliability engineers without prior familiarity with the RPN convention. This is a practical advantage in the academic context of a master's research that aims to articulate mechanical design and reliability analysis.

Table 7: Comparison between the BowTie analysis presented in this work and a conventional FMEA of the same threats.

Aspect	FMEA / RPN	Quantitative BowTie
Causal chain	Implicit	Explicit
Severity assessment	Ordinal (1–10)	Continuous (deterministic)
Pass criterion	RPN threshold	Indicator-by-indicator
Reproducibility	Limited	Direct from sizing
Visualisation	Tabular	Diagrammatic
Identifies critical barrier	No	Yes (via tornado plot)

A concrete numerical illustration of why the conventional RPN collapses information that the BowTie preserves is given in Table 8. The six threats T_1–T_6 of the BowTie diagram were scored on the conventional 1–10 ordinal scales of severity (S), occurrence (O), and detectability (D) following the rating guidelines of the AIAG–VDA 2019 FMEA harmonised handbook [21]: severity is uniformly high because all six threats converge on the same catastrophic top event (valve–piston collision, $S \in [8, 10]$); occurrence is mapped from the

ASTM A877/A877M tolerance band and the catalogued operational variability of the spring (so that occurrence is calibrated to manufacturing rather than to subjective judgement); detectability is uniformly low ($D \in [3, 4]$) because the failure modes are mechanical and predominantly silent before the event of failure. Each score is the median of three independent assignments by the authors using the AIAG–VDA rubric. The resulting RPNs fall in a narrow band (96–240) and place the inner-spring fatigue threat (T_4, RPN $= 150$) and the outer-spring coil-bind threat (T_1, RPN $= 144$) within ten points of each other, even though the quantitative-coupled BowTie identifies both as concurrently at risk through entirely different physical mechanisms. A triaging team using a conventional 100–125 RPN threshold would not be able to distinguish reliably between these two threats. The RPN ranking, in other words, masks the very finding that the BowTie analysis surfaces. The observation is structural rather than specific to the illustrative scores adopted here: because all six threats converge on the same catastrophic top event (valve–piston collision), the severity dimension is saturated and the multiplicative $S \times O \times D$ score loses its discriminating power. The well-documented weaknesses of the multiplicative RPN — non-uniqueness of identical scores, sensitivity to the ordinal scale, and compression of independent risk dimensions into a single number — have been the subject of an extensive critique [22, 23], and the present case study reproduces them in concentrated form. Recent extensions of FMEA — the action-priority logic introduced in the AIAG–VDA 2019 handbook itself, fuzzy-set formulations [23, 24], and AHP-weighted scoring — improve the resolution of the prioritisation but do not address the underlying compression of the causal chain into a single ordinal score, which is the structural objection addressed by the present BowTie formulation.

The two methods are not exclusive: FMEA remains useful for triaging a large number of failure modes when probabilistic data is unavailable, and BowTie is most valuable when a small number of high-severity events deserves a detailed barrier analysis. For the high-performance cylinder head studied here, the BowTie approach is preferred because the failure modes converge on a single top event whose consequences are catastrophic.

A parallel branch of the literature has extended the conventional FMEA through fuzzy-set formulations and AHP-based weighting to address the loss of resolution of the multiplicative RPN. Compared with these approaches, the proposed quantitative-coupled BowTie does not refine the FMEA score; it *replaces* the single-score logic with a barrier-by-barrier deterministic check that is reproducible from the sizing calculations. The two routes are therefore complementary: fuzzy/AHP-FMEA improves the prioritisation among many failure modes, while the present BowTie improves the auditability of the

Table 8: Illustrative AIAG–VDA 2019 scoring of the six threats T_1–T_6 of the BowTie diagram. Severity, occurrence, and detectability follow the harmonised severity-occurrence-detection rubric of Automotive Industry Action Group (AIAG) and Verband der Automobilindustrie (VDA) [21]; each score is the median of three independent assignments by the present authors and *not* the output of a formal expert panel. Occurrence is calibrated to the ASTM A877/A877M tolerance band rather than to subjective frequency judgement. The scores are intended to support a structural comparison between BowTie and RPN, not to constitute a stand-alone risk ranking.

Threat	S	O	D	RPN
T_1 Insufficient spring stiffness	9	4	4	144
T_2 Excessive moving mass	9	3	4	108
T_3 Aggressive cam profile	10	3	4	120
T_4 Spring heating in operation	10	5	3	150
T_5 Out-of-tolerance thermal clearance	8	4	3	96
T_6 Cam-harmonic / spring resonance	10	6	4	240

design margins around a single critical event.

5.3. Limitations

The present analysis has the following limitations, which should be considered when interpreting the results. They are listed in order of priority for follow-on work.

- *No experimental validation — primary limitation.* The quantitative indicators of Table 4 are computed from the analytical sizing of the valve and the spring; no dynamometer, motored-engine, or spring-rig test of the high-performance cylinder head has been conducted. The qualitative validation against four published failure cases reported in Section 4.7 supports the diagnostic behaviour of the procedure but does *not* confirm the absolute magnitude of the indicator margins for the AP 1.8 head. An empirical campaign is required, and its scope is now stated quantitatively for transparency. The minimum experimental verification programme that would close this limitation comprises: (i) a dedicated spring rig to measure the surge response of both springs across the 50–600 Hz band, with laser-displacement instrumentation on the active coils, sufficient to confirm $f_{n,\mathrm{out}} \approx 318$ Hz and $f_{n,\mathrm{in}} \approx 571$ Hz to within $\pm 5\,\%$ under the seat-preload boundary condition; (ii) a motored-engine speed sweep from 6 000 to 10 000 rpm with a non-contacting valve-position transducer sampled at 50 kHz, to verify the absence of valve float at the deceleration peak ($a_{\max,-} = 20\,362$ m s^{-2}) and to measure the residual coil-bind margin within ±0.1 mm; and (iii) a fatigue campaign on a representative inner-spring specimen at the alternating shear stress of $\tau_{a,\text{in}} \approx 374$ MPa superimposed on the mean $\tau_{m,\text{in}} \approx 858$ MPa, run to either run-out at 10^8 cycles or first-failure indication, on a sample size of at least $n = 5$ to support a statistically meaningful estimate of the Goodman factor of safety against the $n_{f,\text{in}} \approx 0.68$ analytical prediction. The total test-rig hours required for this minimum programme are estimated at the order of 600–800 h. The validation roadmap is presented in detail in Section 6.

- *Operating envelope partially characterised.* The deterministic pass criteria are evaluated at the maximum design speed of 10 000 rpm, and Section 4.6 extends the two speed-sensitive indicators (R and f_n/f_{cam}) across 6 000–12 000 rpm at fixed geometry. Off-design behaviour involving thermal transients (cold start, sustained high-coolant-temperature operation) and load transients (part-load, overspeed beyond 12 000 rpm) is not considered, and a coupled thermal–mechanical sweep is identified as a follow-on analytical study.

- *Probabilistic refinement covers manufacturing tolerance but not full operational variability.* The deterministic indicators have been complemented by a Monte Carlo simulation (Section 4.4) propagating the catalogued ASTM A877/A877M spring-wire tolerances and the strength dispersion of Budynas and Nisbett [10, § 10-30] through the analytical sensitivity coefficients. The simulation confirms that the inner-spring Goodman deficit is present in 100 % of the manufactured population, the outer-spring coil-bind margin fails its criterion in 77.9 % of trials with a 5 % probability of actual hard bind, and the inner-spring coil-bind margin is re-classified from active (deterministic) to marginal (P[fail] = 16.2 %). However, the Monte Carlo treats only the manufacturing tolerance and the strength scatter; operational variability beyond ambient temperature (thermal soak transients, oil-condition degradation, lubricant contamination, valve-seat wear) and the cam-profile harmonics distribution are not propagated. A second-generation analysis that extends the simulation to those operational sources and computes joint barrier-failure probabilities (e.g. correlated coil-bind and Goodman events under thermally degraded conditions) is identified as a follow-on direction.

- *Hot clearance treated as a 1-D differential estimate.* The hot-clearance indicator $c_{\mathrm{cold,req}} = 1.25$ mm (bracket [1.13, 1.60]) of Section 4.2 is a closed-form differential of the steel-stem and aluminium-head expansion paths along the load-bearing line that closes the rocker–valve-tip clearance. It relies on a documented stem temperature $\Delta T_{\mathrm{stem}} \approx 600$ K [19], a head temperature $\Delta T_{\mathrm{head}} \approx 150$ K [1], and a generic OHC head path length $L_{\mathrm{Al}} \approx 100$ mm. A coupled thermal–structural finite-element analysis of the specific AP 1.8 cylinder head would refine L_{Al} and the localised ΔT_{head} field and tighten the [1.13, 1.60] mm bracket, but the present deterministic indicator is no longer a qualitative verification.

- *Barrier set restricted to the spring–cam dynamic interaction.* The qualitative validation of Section 4.7 flagged two failure mechanisms that are not currently covered by the preventive set: out-of-tolerance valve-seat alignment [5] and lubrication-regime degradation in the camshaft bearing pair [7]. Candidate indicators for these two mechanisms — a dimensionless stem-to-seat runout ratio ρ_{run} and a Stribeck-curve minimum film-thickness ratio Λ_{min}, both with literature-anchored pass criteria — are specified in Section 6.2 and identified as priority directions for future work to broaden the coverage of valve-train-related failure mechanisms.

- *Single design instance.* The analysis is conducted on a single bespoke cylinder head ($n = 1$). The methodology itself is transposable to other high-rpm SI heads, and Section 4.6 demonstrates the procedure across the operating envelope 6 000–12 000 rpm for the same head, recovering in closed form the surge, dynamic, and loss-of-contact boundaries. The numerical findings (critical barrier identity, indicator margins) at the 10 000 rpm design point are nevertheless case-specific and should not be generalised to other heads without re-evaluation of the geometric and material inputs.

- *Out of scope by design choice.* Knock and other abnormal combustion phenomena, low-speed pre-ignition, IoT-based monitoring, artificial-intelligence-supported diagnostics, and the multi-objective optimisation of the cylinder-head geometry are explicitly out of scope. The reader is referred to the underlying dissertation [3] for the air-flow and combustion analysis of the same head.

6. Validation roadmap and future work

The deterministic indicators of Table 4 and the Monte Carlo probabilities of Table 5 establish a design-stage diagnosis but do not by themselves substitute for the empirical confirmation of the predicted margins. This section specifies (i) the minimum test-rig programme that would close the empirical gap on the AP 1.8 head, (ii) the candidate barrier extensions motivated by the construct-validity benchmark of Section 4.7, and (iii) the analytical and optimisation directions that follow.

6.1. Test-rig programme with statistical acceptance criteria

The minimum experimental verification programme that would close the empirical gap on the analytical findings for the AP 1.8 head comprises three coordinated stages. Each stage is specified with an explicit sample size, instrumentation envelope, and statistical acceptance rule keyed to the analytical predictions of Section 4.2 and the probabilistic envelope of Section 4.4.

1. **Spring rig — modal verification ($n \geq 3$ specimens per spring).** A single-spring test fixture instrumented with a non-contacting laser-displacement sensor (≥ 1 kHz sampling) on the active coils identifies the surge response across the 50–600 Hz band, under the catalogued seat-preload boundary condition. *Acceptance criterion*: the sample-mean first surge frequency $\bar{f}_n$ from $n \geq 3$ specimens shall fall within the 90 % confidence interval of the analytical prediction (calibrated to the MC distribution of Section 4.4, i.e. $f_{n,\text{out}} \in [304, 334]$ Hz and $f_{n,\text{in}} \in [545, 600]$ Hz at 90 % confidence). A failure of containment is reported as a falsification of the analytical model and triggers re-evaluation of the active-mass hypothesis (total versus Rayleigh-effective).
2. **Motored-engine speed sweep — coil-bind and follower-contact verification.** A motored-engine campaign from 6 000 to 12 000 rpm engine speed (the dynamic-loss-of-contact boundary identified in Section 4.6) is conducted with a non-contacting valve-position transducer sampled at ≥ 50 kHz to resolve the deceleration peak of the cam profile. Two acceptance criteria are applied: *(2a)* absence of valve float at the deceleration peak ($a_{\max,-} = 20\,362$ m s^{-2}, Section 4.2) verified by zero-crossing analysis of the (measured−predicted) lift residual at each rpm point, with a detection threshold of ≥ 0.05 mm departure for ≥ 3 consecutive cam events; *(2b)* the residual coil-bind margin measured at full lift ($n \geq 5$ measurements at 10 000 rpm) shall lie within ± 0.1 mm of the MC 50 % credible interval $M_{\text{out}} \in [0.51, 1.10]$ mm for the baseline outer spring (or $M_{\text{out}} \approx 2.0$ mm post-correction). A single

instance of $M_{\text{out}} < 0$ mm (hard contact) at any rpm within the safe envelope falsifies the corrected design.

3. **Inner-spring fatigue campaign — Goodman criterion verification ($n \geq 5$ specimens).** A representative sample of inner springs is cycled under shear-stress amplitudes reproducing the analytical $\tau_{m,\text{in}} = 858$ MPa and $\tau_{a,\text{in}} = 374$ MPa, with each specimen run to either run-out at 10^8 cycles or first-failure indication. *Acceptance criterion*: with $n = 5$ specimens and assuming a Weibull failure model, the upper one-sided 90 % confidence bound on the empirical Goodman factor of safety $\hat{n}_{f,\text{in}}$ shall remain below unity (≤ 0.78 upper bound for 0 of 5 specimens reaching run-out), thereby confirming the analytical prediction $n_{f,\text{in}} \approx 0.68$ and the MC 5 % quantile $n_{f,\text{in},5\%} = 0.637$. A pass condition ($\hat{n}_f \geq 1$ upper bound) on 5 of 5 specimens reaching run-out would falsify the predicted Goodman deficit and require re-evaluation of the shear endurance limit calibration.

The total test-rig hours for this minimum programme are estimated at the order of 600–800 h.

6.2. Candidate barrier extensions

The construct-validity benchmark of Section 4.7 flagged two failure mechanisms that are not currently covered by the preventive set P_1–P_5. For each, a candidate indicator with a literature-anchored pass criterion is proposed below so that the framework can be expanded without re-architecting the bow diagram.

Soffritti — assembly/tolerance barrier (candidate P_6).. The case of Soffritti et al. [5] attributes premature wear at the rocker-arm/pushrod and rocker-arm/valve interfaces to valve misalignment with respect to the seat insert. A candidate indicator is the dimensionless *stem-to-seat runout ratio*

$$\rho_{\text{run}} = \frac{e_{\text{stem}}}{d_{\text{stem}}}, \tag{9}$$

where e_{stem} is the measured lateral offset of the valve-stem axis from the seat-insert axis at the assembled state and d_{stem} is the valve-stem diameter. A literature-anchored pass criterion of $\rho_{\text{run}} \leq 5 \times 10^{-3}$ is proposed, consistent with the typical 25 μm alignment tolerance reported for production valve trains [19] normalised by a nominal $d_{\text{stem}} = 5$–7 mm. The indicator is computable from a coordinate-measuring-machine inspection of the assembled head and can be added to Table 3 once the inspection protocol is documented. The barrier is not evaluated for the AP 1.8 head in the present work but the indicator form demonstrates that the framework is extensible to assembly-driven threats.

Zhao — lubrication-regime barrier (candidate P_7).. The case of Zhao et al. [7] attributes the camshaft bearing failure to an asperity-contact regime that violates the design assumption of full hydrodynamic lubrication. A candidate indicator is the Stribeck-curve minimum film-thickness ratio

$$\Lambda_{\min} = \frac{h_{\min}}{\sqrt{R_{q,1}^2 + R_{q,2}^2}}, \tag{10}$$

where $h_{\min}$ is the predicted minimum oil-film thickness from the elastohydrodynamic-lubrication (EHL) calculation of the cam–follower contact and $R_{q,1}$, $R_{q,2}$ are the root-mean-square surface roughnesses of the two contacting surfaces. The literature-anchored pass criterion is $\Lambda_{\min} \geq 1.5$ for predominantly hydrodynamic operation, with the asperity-contact regime conventionally identified as $\Lambda_{\min} < 1$ [10, § 12-13]. As with the assembly-tolerance barrier, the indicator is not evaluated for the AP 1.8 head here but its specification closes the gap identified by the construct-validity benchmark and broadens the coverage of the procedure beyond the spring–cam dynamic interaction.

6.3. Subsequent analytical extensions

Subsequent extensions of the work will address (i) a second-generation Monte Carlo propagation that adds correlated operational sources (thermal-soak transients, oil condition, valve-seat wear) and joint barrier-failure probabilities; (ii) a coupled thermal–structural finite-element analysis of the cylinder head to refine the differential hot-clearance estimate of Section 4.2 and tighten the $c_{\text{cold,req}} \in [1.13, 1.60]$ mm bracket; and (iii) the integration of the resulting indicator set as a constraint in the multi-objective optimisation of the cylinder-head geometry. These extensions are sequenced behind the test-rig programme of Section 6.1: until the indicator magnitudes for the AP 1.8 head are confirmed experimentally, the analytical refinements rest on assumptions that the rig data alone can substantiate.

7. Conclusion

This work proposed a quantitative-coupled BowTie risk analysis for the valve train of a high-performance VW AP 1.8 cylinder head operating at 10 000 rpm. The qualitative bow diagram was enriched with five deterministic indicators anchored to literature-based pass criteria, and the indicators were propagated through a Monte Carlo refinement under ASTM A877/A877M wire tolerances and the strength dispersion of Budynas and Nisbett [10, § 10-30] ($N = 2 \times 10^5$ trials). Three findings result: (i) two concurrent at-risk preventive barriers — the outer-spring coil-bind margin ($M_{\text{out}} = 0.78$ mm,

$P[\text{fail}] = 77.9\,\%$) and the inner-spring Goodman factor of safety ($n_{f,\text{in}} \approx 0.68$, $P[\text{fail}] = 100\,\%$); (ii) a residual concern on the outer-spring natural-frequency margin ($f_{n,\text{out}}/f_{\text{cam,max}} \approx 3.81$ against the conventional $4\times$ rule of thumb); and (iii) the procedure identifies multiple concurrent critical barriers that a conventional AIAG–VDA scored FMEA collapses into a narrow RPN band. The empirical confirmation of these analytical findings on the AP 1.8 head, together with the extension of the barrier set to assembly tolerance and lubrication regime motivated by the construct-validity benchmark, is the subject of the validation roadmap of Section 6.

CRediT authorship contribution statement

Darlan Costa Porto: Conceptualization, Methodology, Formal analysis, Investigation, Visualization, Writing – original draft.
Heitor Oliveira Gonçalves: Methodology, Validation, Formal analysis, Writing – review & editing, Supervision.
Florian Pradelle: Methodology, Validation, Writing – review & editing.
José Cristiano Pereira: Conceptualization, Methodology, Supervision, Writing – review & editing, Project administration.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No new experimental data were generated in this study. All analytical inputs of Tables 2 and 4 are fully traceable to the closed-form equations given in Section 4 and to the underlying dissertation [3] (forthcoming); no proprietary data are used. The Monte Carlo script that reproduces Table 5 (random seed 20260519, 200 000 trials, Python 3.12, NumPy 2.2.6) is provided as supplementary material so that the probabilistic refinement of Section 4.4 can be re-executed by any reader. Intermediate sizing data and the source

`.tex`/`.bib` files of the manuscript are available from the corresponding author on reasonable request. The experimental validation roadmap of Section 6 specifies the empirical campaign required to confirm the deterministic indicators reported here.

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